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Are virtual places perceived as real? Belief in equivalence of virtual and physical spaces and its role in shaping willingness to migrate to virtual worlds

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A B S T R A C T

Have virtual spaces become so integrated into daily life that they are now perceived as 'real'? Although philosophical debates have long examined the boundaries between real and virtual places, empirical studies on how people perceive these distinctions remain limited. In three cross-sectional studies conducted in the UK and Poland ($N_{total} = 2360$), we investigated the extent to which virtual places are perceived as equivalent to real-world locations. Our findings revealed that most adults still perceive a strong distinction between virtual and physical places, whereas adolescents reported a higher perceived equivalence. In all three studies, stronger attachment to virtual places predicted higher perceived equivalence, which in turn partially mediated the relationship between attachment and a willingness to move more activities to digital places. We argue that perceived equivalence helps explain how people redefine "place" as technology rapidly reshapes our environments.

1. Introduction

Philosophical debates continue over whether virtual places can be considered as real as physical locations (e.g., Chalmers, 2022). While many studies suggest that people still distinguish clearly between digital and physical spaces Oleksy et al. (2023b); Kumar & Shankar, 2024), the increasing role of virtual environments in everyday life raises the question of whether these boundaries are as rigid as they once were.

In 2025, over five billion people worldwide use social media, spending on average 2 h and 23 min per day on these platforms. Among users aged 16–24, this rises to more than 3 h daily (Kemp, 2025). Nearly one in five internet users report recent experiences with virtual reality technologies, and the global gaming audience now exceeds 3.5 billion. In the U.S., 85 % of teenagers play video games, with 41 % doing so daily, and nearly half report being online "almost constantly" (Gottfried & Sidoti, 2024). From social interactions on social media to virtual tourism and entertainment through games and VR, digital spaces increasingly fulfill a role in people's lives once reserved for the physical world. This shift may be changing how people define what counts as a "real" place.

However, no empirical research directly examines whether—and to what extent—people perceive virtual and real places as equivalent. To fill this gap, we introduce the concept of *perceived equivalence*—a belief about whether virtual and real-world places are equally meaningful and serve the same fundamental functions as real-world locations. This expands previous research on the similarities between physical and digital environments, which has largely focused on the phenomenon of presence— an experience of "being there" in a digital space, as if physically present (Slater et al., 2022; Weber et al., 2021). While presence is tied to momentary experiences, perceived equivalence reflects a more stable belief shaped by personal experience, cultural norms, and societal attitudes toward digital environments. We believe that this concept will provide a better understanding of how people's views on places evolve with technological progress and social change.

We propose that perceived equivalence may be linked to personal experiences with virtual places and may also have implications for everyday choices and behaviors related to spending time in virtual or physical environments. Building on this idea, our study has three main objectives. First, we examine whether people perceive virtual places as truly equivalent to real-world locations. Second, we investigate whether place attachment —the emotional bond individuals form with physical or virtual places that fulfill their needs (e.g. self-esteem or a sense of belonging) —enhances perceptions of equivalence between virtual and physical places. Third, we test whether those who see virtual places as equivalent to real locations are more willing to move more aspects of their lives (e.g., work, entertainment, social life) into digital environments. We check willingness to this shift in two contexts: a prospective metaverse scenario and existing virtual spaces (like games or social media). To address these questions, we conducted two large-scale, nationally representative studies in the United Kingdom and Poland and

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one study among Polish adolescents.

In this research, we define virtual worlds as digital places meaningful to users, varying in form and function. Single-player games have specific goals, VR chats enable exploration and co-creation, discussion groups connect like-minded individuals, and the futuristic metaverse projects aim for full real-virtual integration (Dwivedi et al., 2022). Meanwhile, social media's simple graphics can suffice for those who see these platforms as digital homes or public squares (Shanley & Meacham, 2024).

The paper begins by reviewing the philosophical debate over the reality of virtual experiences. We then distinguish our concept of perceived equivalence from presence, followed by a review of relevant empirical findings. Finally, we present our hypotheses linking place attachment to perceived equivalence, and perceived equivalence to the willingness to shift more of life into virtual environments.

2. The philosophical debate on the equivalence of virtual and real experiences

The question of whether virtual experiences are less valuable than real-world ones is longstanding, but has become more relevant with the rise of immersive technologies (Hindriks & Douven, 2018; Munn & Weijers, 2023). A central philosophical argument against virtual experiences is Robert Nozick's "experience machine" (Nozick, 1974), a hypothetical device that simulates perfect pleasure while disconnecting users from reality. Nozick argued that people would reject such a machine because they value actual engagement with the world and the pursuit of authentic meaning. According to this view, experiences detached from real-world consequences are inherently impoverished and fail to support genuine self-understanding (Dreyfus, 2001; Shea, 2017).

However, not all philosophers share these doubts. Critics point out that Nozick's machine offers illusory, preprogrammed experiences, whereas today's virtual environments—especially those that are social and interactive—enable real agency, self-expression, and meaningful engagement (Chalmers, 2022; Shanley & Meacham, 2024). David Chalmers' theory of virtual realism argues that virtual entities and actions are genuinely real, though digitally mediated. From this perspective, distinctions based purely on ontological status (i.e., "real" vs. "virtual") are misleading. The key difference lies in whether people are conscious agents within these environments. Unlike Nozick's machine, users in VR know they are in a virtual world, retain their memories, and can act meaningfully, making their experiences arguably more authentic and self-relevant (Chalmers, 2022; Silcox, 2019). Additionally, Chalmers and others (e.g., Munn & Weijers, 2023) note that criticisms of virtual experiences must account for the rapid development of VR. Future technologies may simulate sensory input with increasing precision, enhancing the sense of presence and expanding the range of possible experiences. Moreover, virtual worlds offer experiences that sometimes exceed physical constraints: flying, inhabiting alternative bodies, or exploring entirely new realities. These experiences can be especially valuable for people with disabilities, granting access to otherwise unreachable places or actions (e.g., climbing mountains or swimming in coral reefs). As VR technology evolves, its capacity to simulate sensory input and presence is likely to enhance, rather than diminish, the richness of these experiences.

Some critiques of virtual life focus less on its ontological status and more on whether it provides meaningful emotional and relational depth. From a philosophical perspective, genuine human connection is central to a fulfilling life. Yet emerging evidence suggests that virtual spaces increasingly support such connections. Social media platforms, multiplayer games, and shared digital environments enable sustained interaction, collaboration, and emotional intimacy (Pendry & Salvatore, 2015; González-Bailón & Lelkes, 2023; Rzeszewski & Evans, 2020). These connections can be particularly significant when physical environments are inaccessible, unsafe, or unsupportive. Within this context, the idea of the metaverse is often not to replace reality, but to extend it. Technologies like augmented reality (AR) are designed to integrate digital experiences into everyday life, enriching rather than isolating users (Dwivedi et al., 2022; Hadi et al., 2024). This vision contrasts with Nozick's machine by preserving users' awareness of both physical and digital domains and enabling meaningful social interactions across them. In fact, even current digital content—such as memes, livestreams, or digital currencies—already has tangible effects on emotions, political engagement, and social dynamics (Leshkevich & Kataeva, 2023), suggesting that the boundary between virtual and real is already porous and socially impactful.

Concerns regarding the lack of authenticity in virtual environments can also be challenged from the perspective of existentialist philosophy. Thinkers in this tradition argue that authenticity depends not on the medium of experience, but on the individual's ability to act in alignment with their true self. According to Heidegger (1927), authenticity arises when individuals live in accordance with their unique possibilities rather than conforming to societal expectations. If virtual environments offer spaces where people can express their values, pursue meaningful projects, and connect with others on their own terms, they may indeed foster authentic experiences.

In summary, while early critiques like Nozick's frame virtual experiences as inferior, contemporary perspectives highlight how virtual environments differ fundamentally from such static, inauthentic illusions. Virtual worlds may real agency, evolving relationships, and the potential for meaningful and authentic engagement. This ongoing debate invites an empirical question: Do people perceive virtual spaces as equivalent to real locations? Despite extensive philosophical discussion, there is limited empirical research on public perceptions of this equivalence, a gap this study aims to address.

3. Empirical research on the role and status of virtual places

There is currently no research directly examining whether people perceive real and virtual spaces as truly equivalent. However, existing studies examine several areas that may be relevant to understanding how individuals conceptualize the status and role of virtual places. In this section, we briefly review three lines of research that may be relevant to understanding perceived equivalence. First, we discuss presence—the sensory experience of feeling immersed in a virtual space. Second, we present an overview of research on threats related to the development of virtual places. Third, we review research on the functional similarities between physical and digital places, focusing on how people use and engage with them in comparable ways.

3.1. The sense of presence in virtual places

Presence refers to the sensation of being physically present in a virtual environment, as if one were truly there (Barreda-Ángeles & Hartmann, 2022). The literature identifies several types of presence, each linked to different aspects of how users perceive and interact with virtual spaces (Lee, 2004). Spatial presence is the sense of physically being inside a virtual space where surroundings seem tangible. Social presence involves the feeling of sharing a space with others and recognizing them as real partners in interaction. Self-presence refers to identifying with one's avatar—feeling as if the digital body is one's own (Kilteni et al., 2012). The experience of presence is strongly reinforced by technological affordances. Research indicates that advanced immersive systems, such as VR headsets, enhance multiple aspects of presence more effectively than less advanced technologies (Shu et al., 2019) Similarly, different aspects of presence may be activated depending on the medium. For example, video games can more intensively engage spatial presence (Maneuvrier et al., 2020), while social media platforms rely heavily on social presence (Ledbetter & Meisner, 2021; Oh et al., 2023).

As we mentioned above, we argue that presence should be distinguished from perceived equivalence. Presence is a subjective and momentary experience of immersion in a virtual environment, whereas

perceived equivalence reflects a broader, long-term conviction about the extent to which virtual places can function comparably to real-world locations. However, from a theoretical perspective, presence and perceived equivalence may influence each other. A strong sense of presence can make virtual spaces feel more equivalent to physical ones by minimizing bodily cues that signal a difference. Conversely, people who already believe that virtual and real places are equivalent may find it easier to feel immersed, since they are less likely to dismiss virtual spaces as less real or meaningful.

Still, presence alone does not mean that virtual places are seen as truly "real" or as meaningful as physical ones. Cultural attitudes and social concerns can shape how virtual spaces are valued and judged. We review these issues in the next section.

3.2. Perceived threats related to the development of virtual spaces

Existing studies suggest that many people worry about the growing dominance of virtual worlds, echoing Nozick's concerns. For example, Oleksy et al. (2023b) used intergroup threat theory (Stephan et al., 2015) to examine whether perceived threats related to the metaverse influence people's willingness to migrate to virtual spaces. Realistic threats include worries about privacy, surveillance, and losing interest in real-world issues. Symbolic threats involve fears that virtual life may disconnect people from meaningful real-world experiences and their sense of self. This shift might lead individuals to become less equipped to handle real-life challenges, favoring an easier, more joyful, consumerism-driven digital escape (Oleksy et al., 2023b). The results showed that such concerns were expressed by the majority of the respondents in this study.

These concerns indirectly suggest that virtual spaces are still perceived as somewhat "inferior" to real places. In response to the question posed by Chalmers (2022) about whether one can lead a meaningful life in virtual worlds, the majority would likely still respond in the negative. The issue of replacing real places with their virtual counterparts is particularly relevant in the context of tourism research. A recent meta-analysis by Fan et al. (2022) warned that while VR can enhance the tourism experience, it might also reduce the desire to travel in real life. Virtual visits may seem more convenient and immersive, making real travel feel unnecessary. Similarly, during the COVID-19 pandemic, lockdowns and social distancing led many to rely on virtual worlds, weakening their emotional connection to real places and affecting well-being (Cowden et al., 2021).

The above studies highlight concerns that the expanding role of virtual spaces might weaken people's connection to the world and degrade their experiences. However, when individuals have positive experiences with virtual places, such concerns may decrease, potentially leading to a higher perception of equivalence between physical and virtual places.

3.3. Functional aspects of using virtual spaces

Research shows that virtual spaces can serve many of the same functions as physical places—and in some cases, even go beyond them. They often offer opportunities unavailable in the real world. For example, games and virtual chats let users adopt avatars and explore behaviors they might not express offline. This includes developing skills (e.g., social abilities), trying out new perspectives, or expressing their true selves (Freeman et al., 2022; Barreda-Ángeles & Hartmann, 2022). Social media groups also allow people to connect with like-minded others, which is especially valuable when their offline environment lacks support (Ceglarek & Ward, 2016). Virtual environments also make it much easier than physical ones to shape spaces around personal needs. As Boellstorff (2010) notes, place-making is central to digital life. In virtual chats, users can build their own worlds and organize themed events—from LGBTQ support groups and meditation to concerts (Barreda-Ángeles & Hartmann, 2022). Analyzing the virtual world of *Second Life*, Harrison (2009) found that some users tend to create and care for special memorials—such as monuments, buildings, museums, and other heritage sites. This nurturing of a sense of continuity in a virtual place was associated with a greater sense of social cohesion and attachment, which generally enabled the transformation of a virtual location into a meaningful place. Interestingly, users were particularly motivated to care for these places, organizing protests to defend them, and reacting with outrage to any adverse changes (Harrison, 2009; Plunkett, 2011).

At the same time, virtual places reflect some of the same negative dynamics seen in physical spaces. These include defensiveness, intolerance toward differing views (Bail et al., 2018), support for censorship (Ashokkumar et al., 2020), and discrimination—for example, against women in gaming (Choe et al., 2020). The ease of digital migration can also foster ideological echo chambers. One example is the mass shift of right-wing users to fringe platforms with no content moderation, increasing ideological extremism (Atari et al., 2022). In this way, digital migration mirrors physical migration: people often move toward environments that reinforce their existing beliefs.

Taken together, this research shows that despite valid concerns, people increasingly use virtual spaces for socializing, entertainment, and personal growth. They may experience these environments as if physically present and often treat them with the same care as real places—even to the point of defending them from perceived threats. People form emotional attachments and move between virtual and physical settings in similar ways. Functionally, the line between the two is becoming less distinct. Given this, our second research aim was to test whether perceiving virtual places as equivalent to real ones is influenced by positive emotional connections to those places. To measure these connections, we draw on the concept of place attachment from environmental psychology.

4. Relative place attachment as a predictor of perceiving real and virtual environments as equivalent

As demonstrated above, many people perceive virtual places as meaningful, offering social support, opportunities for self-expression, and enhanced entertainment. From the perspective of environmental psychology, such positive attitudes toward places are aspects of *place attachment*—the emotional bonds we form with the places we inhabit (Lewicka, 2010). This attachment can also stem from comparisons between different locations; if one place is seen as more important or better at meeting needs, individuals may become more attached to it than to less fulfilling places. This comparative aspect of place attachment was highlighted by Oleksy, Wnuk, and Lassota (2023a), who examined the extent to which virtual places are perceived as more meaningful or satisfying than real-world places.

However, even if place attachment indicates a deep, positive relationship with virtual environments, it does not necessarily translate into redefining what is "real" and what is "virtual"—that is, whether individuals perceive virtual places as having the same status as physical locations. We argue that stronger place attachment to virtual environments may lead to an increased perception of their equivalence to real-world places for two primary reasons. First, a fundamental role of any place is to meet the basic needs of its residents (Altman & Low, 2012), and an inability to satisfy these needs often results in a negative evaluation of that place (Stokols & Shumaker, 1981). If virtual spaces are perceived to meet these needs more effectively than their real-world counterparts, the qualitative, intuitive boundaries between them should be seen as less significant. Second, although research suggests that experiences in virtual worlds are often viewed as less meaningful than those in the real world (Yamada & Matsuda, 2023), individuals who feel a stronger sense of attachment with virtual environments are more likely to challenge this view. For example, Oleksy et al. (2023b) found that higher attachment to virtual places, compared to real-world locations, negatively correlated with both symbolic and realistic threats

related to the development of virtual worlds.

Therefore, we hypothesize the following:

H1. Greater attachment to virtual places is related to perceiving virtual places as more equivalent to real-world locations.

Further, we examined whether perceiving virtual places as more equivalent to real-world locations would transfer into greater willingness to migrate to virtual places.

5. Perceiving real and virtual environments as equivalent as a predictor of willingness to virtual migration

We propose to understand the willingness to migrate virtually as a readiness to transfer as many activities as possible to virtual environments (Oleksy et al. (2023b); Kumar & Shankar, 2024). Unlike traditional permanent relocations, these virtual migrations emphasize more temporary and flexible movements, allowing individuals to oscillate between virtual and real environments. Previous research has shown that a significant predictor of willingness to migrate virtually is the attachment we feel to both real and virtual places (Oleksy et al. (2023b), Oleksy, Wnuk, & Lassota, 2023a; Kumar & Shankar, 2024).

We aim to test new hypotheses suggesting that perceived equivalence between real and virtual worlds could serve as an important factor mediating the relationship between attachment to virtual places and willingness to migrate virtually. Previous research on real-world migration has shown that cultural similarities and shared values between an individual and the destination country increase willingness to migrate, as these factors reduce adaptation costs and challenges related to moving (Lynch et al., 2022). Similarly, in the context of virtual environments, individuals tend to migrate between platforms and environments that align with their significant characteristics e.g. political views (Atari et al., 2022).

We propose that perceiving virtual worlds as equivalent to real locations introduces a new dimension of similarity beyond cultural and individual characteristics. This equivalence bridges the gap between virtual and real environments, making them more seamlessly interchangeable in the minds of individuals. As a result, people who experience a stronger sense of equivalence may be less likely to perceive their virtual activities as inferior or less significant compared to similar activities and experiences in the physical world. They may also be less likely to view time spent in virtual places as escapism or a disconnection from reality. Therefore, unlike conventional virtual migration drivers (e.g., platform features or supportive communities), perceived equivalence reflects an internal cognitive shift, in which virtual places are not only functional alternatives but are seen as a natural part of the world. Consequently, individuals with a strong belief in equivalence may experience virtual migration as more natural and fulfilling than those who perceive a sharper distinction between virtual and physical spaces. Thus, we hypothesize that:

H2. Perceived equivalence between real and virtual worlds mediates the relationship between place attachment and willingness to migrate to virtual worlds.

6. Overview of the studies

In three online studies—two conducted on adult samples in distinct cultural contexts (the United Kingdom and Poland) and one among Polish adolescents aged 15–18—we examined the perceived equivalence of virtual places to real-world locations and its predictors. Our primary objective was to determine whether individuals perceive virtual places as equivalent to real locations and to explore whether attachment to virtual environments correlates with this perception. We also examined the consequences of perceiving virtual and real places as equivalent, specifically its relationship with individuals' willingness to transfer more activities to virtual spaces (i.e., virtual migration). Study 3 aimed to determine whether the results observed in adult samples align with or differ from those found among younger users who have been more immersed in virtual places from an early age. Additionally, we included a presence scale to empirically test the theoretical distinction between presence and perceived equivalence. All studies were approved by the Scientific Research Ethics Committee at [blinded for review].

7. Study 1

7.1. Participants and procedure

Study 1 was part of a broader research conducted using a nationwide sample of UK citizens (N = 759) representative in terms of age (M = 45.70, SD = 16.69) and gender, men = 369, women = 387, other = 3). In terms of education, 68 % of participants reported having at least a higher education degree, while 32 % did not obtain a college or university degree. Participants were predominantly White (86.8 %), with smaller representation of Asian (7.6 %), Black (3.2 %), Mixed (1.2 %), and other (1.2 %) individuals. Participants assessed how difficult or easy it was for their household to make ends meet, considering the total income from all contributing members, M = 3.06, SD = 1.02 (on a scale of 1 - very difficult to 5 - very easy) The research was conducted by an online research panel Prolific, using computer-assisted web interviewing (CAWI). All participants were compensated monetarily (3 GBP) upon approval of their submissions. The sample size was sufficient for detecting a small-to-medium (0.2–0.5) effect size, assuming a power of 80 % and a probability level of $\alpha = 0.05$.

7.2. Measures

Perceived equivalence between virtual and real place is a new measure developed specifically for this study. Our goal was to capture how strongly individuals cognitively accept virtual places as equivalent to real-world locations, an idea rooted in philosophical perspectives such as Chalmers (2022). We did not create a broad item pool and select a smaller subset; instead, we focused on directly measuring the core concept—the extent to which people see virtual spaces as legitimate equivalents to physical places. We chose four items that directly target this core construct: 1) 'Virtual places are a normal part of the contemporary world', 2) 'Virtual places are just as important as real places in the contemporary world', 3) 'People should not treat virtual places the same way as they treat real locations.' (reverse coded), 4) 'There is no longer any reason to draw a clear line between virtual worlds and reality' ($\alpha = 0.73$). To ensure content validity, all items were reviewed by three experts in environmental psychology and digital media studies. The experts confirmed that the items adequately captured the intended theoretical domain of perceived equivalence between real and virtual places.

Place attachment was measured with relative place attachment scale (used in Oleksy, Wnuk, & Lassota, 2023a). We used four items: 'Some virtual places are more important to me than real-world places.', 'I often feel happier in my favorite virtual world than in the real world.', 'In some virtual places, I feel more at home than I do in the real world.', 'I can be who I wish to be in virtual places to a greater extent than I can in real-world places.'. ($\alpha = 0.88$).

Willingness to migrate to the metaverse was measured with three items from Oleksy, Wnuk, and Lassota (2023a). (1)'I would like to move most of my daily activities to the metaverse.', (2) 'If possible, I would prefer to work in the metaverse rather than in the real world', (3) 'I would rather spend time with friends in the real world than in the metaverse (reverse coded) ($\alpha = 0.63$). Before answering these questions, the users were provided with the following definition of the metaverse:

"The metaverse presents a new vision for the internet, where virtual reality enhances user interactions, offering experiences that closely resemble those in the real world. Through specialized goggles, users can immerse themselves in a digital environment, taking the form of an

"avatar"—an audiovisual representation used to communicate with others. Avatars can take any form, such as a human, robot, or any imagined creature. Designed to provide a range of entertainment options, from sports to games of all kinds, the metaverse enables users to make friends and even date people from around the world, all without leaving home. Businesses can hold virtual meetings with clients, collaborate with employees globally, and offer their services in the metaverse. Users can also shop at virtual malls and attend events and concerts by real-world performers—all within this digital realm".

Social media use was measured with a question: 'How often do you use the following technological tools: 1) social networking sites, 2) discussion groups, and 3) online chats Answers were given on a scale from 1 ('never') to 7 ('very often') and then averaged ($\alpha = 0.68$).

Virtual Reality/Games use was measured with a question: 'How often do you use the following technological tools: 1) single-player games, 2) multiplayer games, 3) virtual reality. Answers were given on a scale from 1 ('never') to 7 ('very often'). Answers were given on a scale from 1 ('never') to 7 ('very often') and then averaged ($\alpha = 0.69$).

Additionally, we measured demographic variables: gender, age, socioeconomic status ethnicity, and the education level.

7.3. Analytical overview

The first step involved assessing the psychometric properties of the Perceived Equivalence scale. Then, we conducted regression analyses with equivalence as a dependent variable. Finally, we conducted mediation analyses to examine whether equivalence mediates the relationship between place attachment and willingness to migrate to metaverse/virtual worlds.

7.4. Results

7.4.1. Construct validity of the perceived equivalence scale

We first conducted a confirmatory factor analysis (CFA) in Mplus (MLR estimation) to test whether the four items form a single latent factor. The model fit was excellent: $\chi^2(2) = 5.43$, p = .0661, RMSEA = 0.05, CFI = 0.99, TLI = 0.98, and SRMR = 0.02. Three items (1, 2, and 4) had high standardized loadings ranging from 0.58 to 0.88. The reverse-coded item (3) produced a slightly lower loading (0.46), but still remained above the accepted threshold of 0.40 (DeVellis & Thorpe, 2021). These results confirm that the four items tap into a single underlying dimension.

Second, we examined the convergent validity of Perceived Equivalence scale. As expected, the *Perceived Equivalence* scale correlated positively with key constructs from our theoretical model, namely place attachment ($r = .50$) and willingness to migrate ($r = .35$). Additional analyses showed that perceived equivalence was also positively associated with the frequency of video game use ($r = .29$, $p < .001$) and social media use ($r = .34$, $p < .001$). These patterns further support the scale's convergent validity, indicating that individuals who spend more time in virtual contexts also view these spaces as more equivalent to real-world places (see Table 1).

To demonstrate the discriminant validity, we confirmed that *Perceived Equivalence* is empirically distinct from other constructs in our model—namely, *Place Attachment* and *Willingness to Migrate*. To do so, we conducted a series of confirmatory factor analyses (CFAs) comparing alternative models. A three-factor model (equivalence, place attachment, and willingness to migrate) showed a significantly better fit (Study 1: $\chi^2(41) = 134.75$, $p < .001$; CFI = 0.96; RMSEA = 0.06; SRMR = 0.03) than a single-factor model forcing all items to load onto one latent dimension ($\chi^2(44) = 594.18$, $p < .001$; CFI = 0.79; RMSEA = 0.11; SRMR = 0.17). This difference indicates that, although these constructs are conceptually related, they remain empirically distinguishable.

Taken together, these results demonstrate that the *Perceived Equivalence* measure offers a unidimensional and psychometrically good assessment of the extent to which people cognitively accept virtual places as on par with real-world locations.

Table 1
Means, standard deviations, and correlations with confidence intervals.

Variable	*M*	*SD*	1	2	3	4
1. Perceived equivalence	2.95	1.19				
2. Place attachment	2.60	1.44	.50** [.44, .55]			
3. Willingness to migrate to the metaverse	2.23	1.23	.35** [.29, .41]	.54** [.49, .59]		
4. VR/Games use	2.66	1.46	.29** [.23, .36]	.44** [.38, .50]	.30** [.23, .36]	
5. Social media use	3.76	1.51	.33** [.27, .40]	.32** [.26, .39]	.18** [.11, .25]	.40** [.34, .46]

Note. * indicates $p < .05$. ** indicates $p < .01$.

Next, we conducted one-sample *t*-test to examine whether the perceived equivalence between real and virtual places was significantly different from the midpoint of the scale. The perceived equivalence was significantly below the midpoint in both the UK and Poland (ps < .001).

7.4.2. Regression analysis

The results of the regression analysis showed that both place attachment and the use of social networking sites predicted the perceived equivalence between real and virtual places, while the use of games and VR did not (see Table 2).

7.4.3. Mediation model

Then, we examined the mediation of place attachment on willingness to virtual migration via perceived equivalence between real and virtual places, using PROCESS macro (v.4.2.) for SPSS, model 4 (Hayes, 2013). In both studies, the indirect effect was statistically significant (IE = .05, SE = .02, 95 % CI [.01.08]). The coefficients of the model are presented in Fig. 1. Adding covariates (age, gender, SNS use, games/VR use, SES, and education level) did not meaningfully change the results.

7.5. Discussion

Study 1 provided support for the psychometric properties of the Perceived Equivalence scale, confirming its unidimensional structure and demonstrating both convergent and discriminant validity. Perceived equivalence was significantly below the midpoint of the scale in both adult samples, suggesting that most participants did not view virtual places as equivalent to physical locations. Place attachment was a strong positive predictor of perceived equivalence, supporting the hypothesis that emotional bonds with virtual places contribute to perceiving them as more equivalent to real-world locations. The mediation effect of perceived equivalence between place attachment and

Table 2
Regression analysis.

DV	Perceived equivalence	
	B	*SE*
Constant		
Place attachment	.33**	.03
SNS use	.12**	.03
Games/VR use	.02	.03
Age	−.01**	.003
Gender (0 – man, 1 – woman)	.14	.07
Education	.05	.04
SES	−.001	.001
R^2	.30	

Note. **$p < 0.001$.

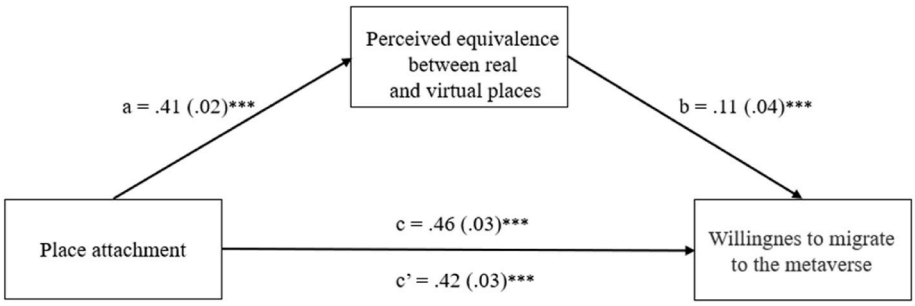


Fig. 1. Effects of attachment to virtual places on willingness to migrate to metaverse via perceived equivalence between real and virtual places. *p < .05, **p < .001. ***p < .0001.

willingness to migrate to virtual spaces was replicated in both studies. Social media use was a weak predictor of perceived equivalence, while games/VR use was not, suggesting that social interaction rather than immersion plays a more important role in shaping perceptions of virtual places as equivalent to the physical locations.

8. Study 2

8.1. Participants and procedure

In Study 2, we aimed at replication of our results in a Polish sample, examining the extent to which participants perceived virtual and real places as equivalent. To increase robustness, we also broadened the context of the virtual migration destination—shifting the focus from the metaverse to virtual worlds more generally. This study was part of a broader research conducted using a nationwide sample of Polish citizens (N = 1055) via an online research panel Ariadna, using computer-assisted web interviewing (CAWI). All participants were rewarded with points for their participation, which could later be exchanged for items available through the platform's reward system. To ensure the representativeness of a sample, a random quota sampling method was employed, considering gender, age, and size of residence. (M_{age} = 47.77, SD = 15.86, men = 502, women = 553). The participants were geographically diverse, with 30 % living in large or very large cities (over 100,000 inhabitants), 33 % residing in small or medium-sized towns, and the remaining participants living in rural areas. Again, the sample size in both studies was sufficient for detecting a small-to-medium (0.2–0.5) effect size, assuming a power of 80 % and a probability level of α = 0.05.

8.2. Measures

Perceived equivalence between virtual and real places was measured using the same scale as in Study 1 (4-item version: α = .68; 3-item version: α = .74). The decision to use the reduced version is discussed in more detail in the Results section.

Willingness to migrate to the virtual worlds was measured with three items based on willingness to migrate to the metaverse (Oleksy, Wnuk, & Lassota, 2023a). (1)'I would like to move most of my daily activities to virtual places.', (2) If only it were possible, I would prefer to spend all my time in virtual places, (3) 'I would rather spend time with friends in virtual places than in the real world (α = 0.94).

Place attachment (α = 0.91), Social media use (Study 2: α = 0.80), and Virtual Reality/Games use (α = 0.71) were.

Additionally, we measured demographic variables: gender, age, size of place of residence, and time spent in virtual places.

8.3. Analytical overview

The analytical strategy followed the same steps as in Study 1.

8.4. Results

8.4.1. Confirmatory analyses

In comparison to Study 1, the single-factor model including all four Perceived Equivalence items showed suboptimal fit in the Polish sample ($\chi 2$ (2) = 63.61, p < .001; CFI = 0.92; RMSEA = 0.17; SRMR = 0.05). Factor loadings analysis indicated that the reduced model fit was primarily driven by one reverse-coded item in the Perceived Equivalence scale (λ = 0.12). Consequently, this item was removed from the final model. The revised model with three items was just-identified (df = 0), which does not allow fit indices. However, all remaining factor loadings were high and statistically significant (λ = 0.68–0.91, p < .001), supporting the internal consistency and construct validity of the scale. Therefore, we used the three-item version of the Perceived Equivalence scale in all subsequent analyses in Study 2, as it maintained theoretical coherence while improving psychometric performance.

Second, we examined the convergent validity of Perceived Equivalence scale. Smilar to Study 1, the perceived equivalence correlated positively with place attachment (r = .60) and willingness to migrate (r = .51). Additional analyses showed that perceived equivalence was also positively associated with the frequency of video game use (r = .35, p < .001) and social media use (r = .43, p < .001) (see Table 3).

Confirmatory factor analysis conducted in MPlus showed that equivalence (measured with three items), place attachment and willingness to migrate to virtual worlds were again empirically distinct constructs. A modes with three latent factors fitted the data well ($\chi 2$ (32) = 192.65, p < .001; CFI = 0.96; RMSEA = 0.07; SRMR = 0.04). All factor loadings were above 0.60. We also run additional factor analysis, in which all items were forced to load one factor. This resulted in a worse model fit ($\chi 2$ (35) = 905.93, p < .001, CFI = 0.80, RMSEA = 0.15, SRMR = 0.08).

Table 3
Means, standard deviations, and correlations with confidence intervals.

Variable	*M*	*SD*	1	2	3	4	5
1. Perceived equivalence	3.53	1.38					
2. Place attachment	2.86	1.55	.60** [.56, .64]				
3. Willingness to migrate to virtual worlds	2.35	1.49	.51** [.46, .55]	.69** [.65, .72]			
4. VR/Games use	2.68	1.52	.35** [.30, .40]	.45** [.40, .50]	.46** [.41, .50]		
5. Social media use	3.45	1.48	.43** [.38, .48]	.41** [.36, .46]	.36** [.31, .41]	.52** [.47, .56]	
6. Time spent in virtual places.	1.61	.87	.24** [.18, .30]	.32** [.26, .37]	.26** [.21, .32]	.36** [.31, .41]	.35** [.29, .40]

Note. * indicates p < .05. ** indicates p < .01.

Next, we conducted one-sample t-tests to examine whether the perceived equivalence between real and virtual places was significantly different from the midpoint of the scale. The perceived equivalence was significantly below the midpoint ($p < .001$).

8.4.2. Regression analysis

The results of the regression analysis 2 showed that both place attachment and the use of social networking sites predicted the perceived equivalence between real and virtual places, while the use of games and VR did not (see Table 4).

8.4.3. Mediation model

Then, we examined the mediation of place attachment on willingness to virtual migration via perceived equivalence between real and virtual places, using PROCESS macro (v.4.2.) for SPSS, model 4 (Hayes, 2013). The indirect effect was statistically significant (IE = .08, SE = .02, 95 % CI [.05; .12]). The coefficients of the model are presented in Fig. 2. Adding covariates (age, gender, size of place of residence, SNS use, games/VR use, and time spent in virtual places) did not meaningfully change the results.

8.5. Discussion

The important difference from Study 1 was that one reverse-coded item performed poorly in the Polish sample, even though it had functioned adequately in the English version. This discrepancy aligns with prior research showing that reverse-worded items can introduce measurement inconsistencies in cross-lingual contexts, owing to factors such as semantic nuances, increased cognitive load, or response styles (Weijters & Baumgartner, 2012; van Sonderen et al., 2013). Once this problematic item was removed, the remaining three items all showed high and statistically significant factor loadings, indicating that the revised scale effectively captured the underlying construct. Notably, excluding this item did not alter the overall pattern of results, which closely mirrored those obtained in Study 1. Place attachment and social media use remained significant predictors of Perceived Equivalence, whereas VR/games use did not. The mediated effect of place attachment on willingness to migrate via perceived equivalence was also replicated.

9. Study 3

9.1. Participants and procedure

In Study 3 ($N = 546$), we investigated the extent to which Polish adolescents aged 15–18 (M = 16.30, SD = 1.08, 276 boys and 270 girls) perceive virtual and real places as equivalent. Research was conducted via an online research panel Ariadna, using computer-assisted web interviewing (CAWI). As the study involved minors, all adolescent participants were part of the research panel with parental consent, ensuring compliance with ethical guidelines. All participants were rewarded with points for their participation, which could later be exchanged for items available through the platform's reward system. The sample was geographically diverse, with 31 % living in large or very large cities (over 100,000 inhabitants), 37 % residing in small or medium-sized towns, and the remaining participants living in rural areas.

Table 4
Regression analysis.

DV	Perceived Equivalence	
	B	*SE*
Constant		
Place attachment	.47**	.03
SNS use	.19**	.03
Games/VR use	.04	.03
Age	.01**	.002
Gender (0 – man, 1 – woman)	−.15	.07
Size of place of residence	−.02	.02
Time spent in virtual places	.04	.04
R^2	.41	

Note. $^{**}p < 0.001$.

9.2. Measures

In Study 3, we measured perceived equivalence (4-item version: α = .61; 3-item version: α = .76), relative place attachment (α = .86), willingness to migrate to virtual worlds (α = .63), social media use (α = .65), and games/VR use (α = .73) with the same measures as in Study 2.

Presence was assessed using the Igroup Presence Questionnaire (IPQ) (Schubert et al., 2001; Polish adaptation: Strojny et al., 2022). The IPQ evaluates the subjective experience of presence in virtual environments and consists of three subscales: spatial presence (e.g., "I have the feeling that the virtual world surrounds me"), involvement (e.g., "I lose awareness of the real world while being in the virtual environment"), and realness (e.g., "The virtual environment seems completely realistic"). Participants responded using a 7-point Likert scale (1 = strongly disagree to 7 = strongly agree), with higher scores indicating a stronger sense of presence. In this study, we adapted the IPQ instructions to fit the context of our research, instructing participants to reflect on the most engaging virtual place they regularly use rather than a specific experimental environment. Factor analysis showed that all items loaded onto a single factor, so following the recommendations of Strojny et al. (2022), we computed an overall presence index by averaging all items (9-item version: $\alpha = .82$, 7-item version: α = .90).

Additionally, we collected demographic variables, including age, gender, size of place of residence, and socioeconomic status.

9.3. Analytical overview

The analytical strategy was the same as in Studies 1 and 2.

9.4. Results

9.4.1. Confirmatory analysis

As in Study 2, the initial confirmatory factor analysis (CFA) for the Perceived Equivalence scale in the Polish adolescent sample revealed suboptimal fit indices ($\chi^2(2) = 51.90$, $p < .001$; RMSEA = .21; CFI = 0.84; SRMR = 0.07). Examination of standardized loadings indicated that the same reverse-coded item from the previous study again exhibited a low loading. Consistent with the findings in Study 2, removing this problematic item left the remaining three items with high and statistically significant loadings (λ = .62–0.85, $p < .001$), suggesting that the reduced version of the scale adequately captured the underlying construct. For this reason, we used the three-item version of the Perceived Equivalence scale in all subsequent analyses.

Second, we examined the pattern of zero-order correlations. Place attachment showed a strong positive association with perceived equivalence between real and virtual places. Both place attachment and equivalence were related to willingness to migrate to the virtual worlds. Presence was not significantly correlated with perceived equivalence and showed a weak negative correlation with place attachment (see Table 5). Next, we conducted a one-sample *t*-test to examine whether the perceived equivalence between real and virtual places was significantly different from the midpoint of the scale. The perceived equivalence was significantly above the midpoint ($p < .001$).

Confirmatory factor analysis conducted in MPlus showed that presence, equivalence, place attachment and willingness to migrate to the metaverse/virtual worlds were empirically distinct constructs. The first model with four latent factors demonstrated a suboptimal but reasonable fit to the data ($\chi2\ (146) = 531.56$, $p < .001$; CFI = 0.89; RMSEA = 0.07; SRMR = 0.07). Factor loadings analysis indicated that the reduced model fit was primarily driven by two reverse-coded items in the Presence scale (λs = 0.007 and 0.20). Removing these items improved model

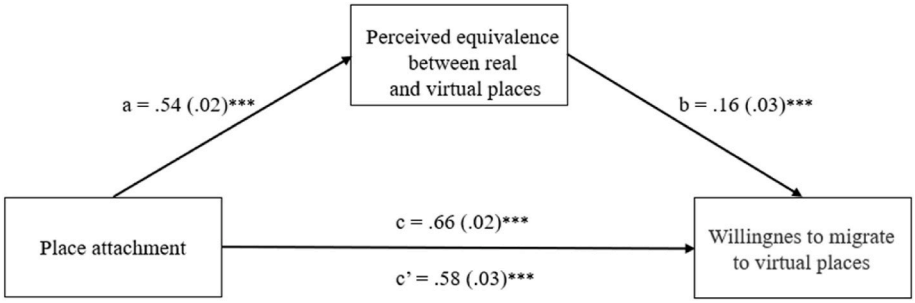


Fig. 2. Effects of attachment to virtual places on willingness to migrate to virtual places via perceived equivalence between real and virtual places. *p < .05, **p < .001. ***p < .0001.

Table 5
Means, standard deviations, and correlations with confidence intervals.

Variable	*M*	*SD*	1	2	3	4	5
1. Perceived equivalence	4.46	1.37					
2. Presence	3.61	1.31	−.05 [-.13, .03]				
3. Place attachment	4.41	1.41	.50** [.43, .56]	−.16** [-.25, −.08]			
4. Willingness to migrate to the virtual places	4.57	1.41	.36** [.29, .43]	−.27** [-.35, −.19]	.58** [.53, .64]		
5. Social media use	4.26	1.25	.06 [-.03, .14]	.36** [.28, .43]	.01 [-.07, .10]	−.11** [-.19, −.03]	
6. VR/Games use	4.39	1.54	.04 [-.04, .13]	.39** [.31, .46]	−.03 [-.12, .05]	−.08 [-.16, .01]	.44** [.37, .50]

Note. * indicates $p < .05$. ** indicates $p < .01$.

fit ($\chi 2$ (113) = 339.97, p < .001; CFI = 0.93; RMSEA = 0.06; SRMR = 0.05). We also run additional factor analysis, in which all items were forced to load one factor. This resulted in a substantially worse model fit ($\chi 2$ (170) = 2444.46, p < .001, CFI = 0.39, RMSEA = 0.16, SRMR = 0.18).

Therefore, we used the seven-item Presence scale in subsequent analyses. However, we also conducted additional analyses (correlations, regressions, and mediation) using the full four-item Equivalence scale and the original nine-item Presence scale, including the reverse-coded items. The results remained largely unchanged, confirming that the inclusion or exclusion of these items did not substantially affect the overall findings.

9.4.2. Regression analysis

The results of the regression analysis showed that place attachment predicted perceived equivalence, while the SNS use, games/VR use and presence did not (see Table 6).

9.4.3. Mediation model

Then, we examined the mediation of place attachment on willingness to virtual migration via perceived equivalence between real and virtual places, using PROCESS macro (v.4.2.) for SPSS, model 4 (Hayes, 2013). The indirect effect was statistically significant (IE = .05, SE = .02, 95 % CI [.004; .09]). The coefficients of the model are presented in Fig. 3. Adding covariates (age, gender, SES, size of place of residence, SNS use, and games/VR use) and presence did not meaningfully change the results.

Table 6
Regression analysis.

DV	Perceived equivalence	
	B	*SE*
Constant		
Place attachment	.49**	.04
Presence	.01	.04
SNS use	.03	.05
Games/VR use	.04	.04
Age	.02	.05
Gender (0 – girl, 1 –boy)	.03	.11
SES	.04	.06
Size of place of residence	−.01	.04
R^2	.25	

Note. **$p < 0.001$.

9.5. Discussion

We found that, unlike in adult samples, perceived equivalence among adolescents was significantly above the midpoint of the scale. A similar effect was observed for relative place attachment, which was also higher among adolescents. Importantly, the results confirmed that perceived equivalence is a distinct construct from presence, with no correlation between them. The mediation effect of perceived equivalence between place attachment and willingness to migrate to virtual worlds was replicated, remaining robust even when presence and other covariates were included.

As in Study 2, the reverse-coded item in the Perceived Equivalence scale performed poorly and had to be excluded. This pattern, also observed for a reverse-coded item in the validated IPQ Presence Scale, highlights a recurring issue with reverse-worded items in Polish-language surveys. Importantly, removing the problematic item did not alter the overall pattern of results.

10. General discussion

The debate on the nature of reality and the authenticity of experiences has long been central to philosophy (for a review, see Chalmers, 2022). With the development of digital spaces and virtual reality, questions about the value of virtual experiences have become increasingly relevant. Is spending time in virtual spaces as meaningful as in the physical world? Can virtual exploration be as fulfilling as real-life journeys? From Nozick's unsettling vision of the experience machine to Chalmers' concept of virtual realism, the question of whether digital places and the experiences they offer can be equated with those in the real world remains fundamental. Our study examined the prevalence of the belief in the equivalence between virtual and real places as well as its predictors and implications for virtual migration readiness.

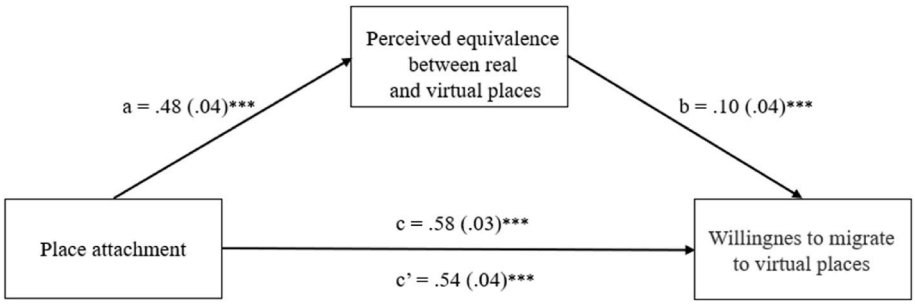


Fig. 3. Effects of attachment to virtual places on willingness to migrate to virtual places via perceived equivalence between real and virtual places (Study 3). *p < .05, **p < .001. ***p < .0001.

10.1. Perceived equivalence between real and virtual places

Using two representative samples from the UK and Poland, we examined how adults perceive the status of virtual places relative to physical locations. In a third study, we explored whether these perceptions differ among adolescents in Poland. Among adult participants, the perceived equivalence between virtual and real places was significantly below the midpoint of the scale in both countries, suggesting that most did not view virtual spaces as comparable to physical ones. These findings stand in contrast to Chalmers' (2022) concept of virtual realism, which posits that digitally mediated environments and actions hold genuine meaning, much like their physical counterparts. Our results indicate that this philosophical perspective has not yet translated into widespread public perception—at least among adults. However, the results obtained among adolescents in the Polish sample differed from both adult samples—their perceived equivalence scores were higher than in case of adults.

The low level of perceived equivalence among adults may reflect a more traditional view of reality, in which physical places are seen as fundamentally distinct from virtual ones. It may also arise from broader public concerns about the expanding role of virtual environments. These concerns may explain why the public perceives virtual spaces as distinct from real places despite their growing significance. The increasing influence of these worlds raises realistic threats, such as excessive control by tech corporations and loss of privacy, as well as symbolic threats, whereby virtual experiences are perceived as distancing us from inherently human experiences (Oleksy et al. (2023b); Kumar & Shankar, 2024). These concerns are echoed by Haidt (2024), who argues that the digitalization of daily life poses significant risks to the development and mental health of children and adolescents (see also Orben & Blakemore, 2023). Thus, the observed gap between Chalmers' theory and public perception may be driven, at least in part, by broader societal anxieties surrounding the "invasion" of virtual spaces into daily life. Future studies should directly examine the roles of threats related to excessive digitalization and the importance of physicality in shaping the perceived equivalence between real and virtual places.

On the other hand, the findings in the adolescent sample may reflect the more central role that digital spaces play in young people's daily lives (Soh et al., 2024), both in terms of time spent online and the depth of engagement. Surveys indicate that over 35 % of teenagers report using platforms such as TikTok, YouTube, Instagram, and Snapchat "almost constantly" (Vogels et al., 2022), with adolescents consistently engaging more with digital media than older generations (Wike et al., 2022). While most adolescents are online daily, older adults vary widely in their usage, though generational gaps are narrowing. Adolescents themselves often act as digital brokers, helping their parents navigate online tools and platforms (McKenzie et al., 2022).

More than screen time, however, it is the significance of these environments that may explain adolescents' higher perception of equivalence. Their digital engagement is qualitatively different, often playing a formative role in identity development. According to Bronfenbrenner's bioecological theory, digital spaces are not merely extensions of the physical world but key developmental contexts for adolescents (Navarro & Tudge, 2023). They operate in overlapping microsystems, moving fluidly between online and offline worlds, which may normalize virtual environments as equally meaningful. Virtual spaces also serve as identity playgrounds. Reduced social cues, asynchronicity, and platform-specific norms allow adolescents to experiment with different aspects of self (Manago & McKenzie, 2022). For example, identity expression on Facebook tends to reflect social conformity, while anonymous platforms invite more personal or experimental self-presentations (Livingston & Sefton-Green, 2016). Moreover, adolescents are not merely passive participants in digital spaces; they frequently act as active co-creators, shaping dynamic digital norms and subcultures, as well as inviting others to participate (Subrahmanyam & Smahel, 2010).

The role of virtual spaces in adolescent identity formation may foster a stronger sense of place identity in relation to digital environments. Place identity refers to how individuals define themselves through their connection to a place—how it reflects their values, supports continuity, and shapes their sense of self (Twigger-Ross & Uzzell, 1996). Our results support this view: adolescents scored higher on the relative place attachment scale, which incorporates aspects of place identity in comparing virtual and physical spaces. The deep integration of digital environments into their daily lives may help explain their higher levels of perceived equivalence.

In summary, these findings offer an empirical contribution to long-standing philosophical debates. Nozick's "experience machine" argued that simulated experiences lack the depth and value of those grounded in physical reality. In contrast, Chalmers (2022) proposes that modern virtual environments can support meaningful, authentic experiences. Our data suggest a generational divide in alignment with these positions: older adults echo Nozick, seeing virtual spaces as fundamentally distinct, while adolescents resonate more with Chalmers' virtual realism, perceiving virtual and physical places as increasingly equivalent.

These generational differences likely reflect broader psychological and social shifts associated with the rise of "digital natives"—a generation raised with near-constant access to virtual spaces (e.g., social media). This intensity of engagement, visible in global usage patterns (Gottfried & Sidoti, 2024; Kemp, 2025) may normalize virtual experiences to the point where they are perceived not only as meaningful but also as ontologically "real." Our finding that adolescents perceive virtual and physical spaces as highly equivalent raises important societal concerns. Research highlights that Generation Z experiences increased anxiety, loneliness, and declining levels of in-person social interaction, life satisfaction, and happiness (Twenge, 2017; Twenge & Blanchflower, 2025). Many of these effects have been attributed to changing communication patterns and the displacement of offline activities (Twenge & Spitzberg, 2020; Haidt, 2024). If younger generations continue to treat digital environments as equally real and meaningful, this may further reduce their engagement with in-person communities and intensify withdrawal from offline social contexts. These "virtual

migrations" could deepen mental health challenges and undermine social cohesion (Haidt, 2024).

10.2. Distinguishing perceived equivalence of real - virtual places from sense of presence in virtual places

Our research moves beyond prior studies that focused on similarities between virtual and physical places through the lens of presence (Slater et al., 2022; Weber et al., 2021). We conceptualized perceived equivalence as a broader evaluation of the status of virtual places in society, in contrast to presence, which captures the immediate feeling of "being there" in a virtual environment. Notably, Study 3 found no correlation between these constructs. Controlling for presence also did not affect the mediation model linking place attachment to virtual migration via perceived equivalence. These findings highlight the conceptual distinction between presence and perceived equivalence. However, our use of a cross-sectional design and general self-report format has limitations. Participants rated their sense of presence based on their most engaging virtual environment, while presence is typically assessed in the context of a specific platform or moment (e.g., VR chats or lab-based simulations; Barreda-Ángeles & Hartmann, 2022; Strojny et al., 2022). Future studies should examine the relationship between these constructs using more conventional presence measures, targeted at specific platforms. They should also test whether similar patterns hold among adults and explore these dynamics in experimental and longitudinal settings—for example, by investigating whether perceived equivalence moderates the ability to experience presence in immersive environments.

10.3. The role of place attachment in increasing perceived equivalence between virtual and real-world places

We hypothesized that stronger attachment to virtual places would predict greater perceived equivalence between virtual and real-world locations. Across all three studies, place attachment to virtual environments significantly predicted perceived equivalence, confirming Hypothesis 1. These findings align with environmental psychology, which distinguishes between "spaces" and "places"—the latter defined by their emotional and functional significance (Tuan, 1979; Wnuk et al., 2021). A space becomes a place when it fulfills individuals' goals and supports identity and belonging. Our results suggest that this process operates similarly in digital environments: when virtual places meet psychological needs, they are more likely to be perceived as equivalent to real ones.

Stronger attachment to virtual environments may also reduce perceived threats linked to virtualization, shifting focus from differences to similarities between real and virtual places (Oleksy et al. (2023b)). This interpretation is consistent with Relph's (1976) phenomenological perspective, which emphasizes the experiential and emotional basis of place over physical attributes (see also Seamon, 2012). Importantly, we observed generational differences in virtual place attachment. Adult participants in both countries generally reported lower attachment to virtual spaces, likely reflecting a stronger connection to physical places. Adolescents, in contrast, showed higher attachment to digital environments, suggesting that growing up in technologized settings fosters a deeper personal connection to virtual spaces (Manago & McKenzie, 2022). As a result, they may be more inclined to see them as equally meaningful and "real."

10.4. Perceived equivalence between real and virtual worlds as a mediator between place attachment and willingness to migrate virtually

Does perceiving virtual places as equivalent to real ones influence willingness to migrate virtually? Our results confirm Hypothesis 2, showing that perceived equivalence partially mediates the relationship between place attachment and willingness to transfer activities to virtual spaces. This suggests that equivalence perceptions help reduce psychological, emotional, and cognitive barriers to virtual migration. Just as cultural and value similarity between countries facilitates real-world migration (Lynch et al., 2022), perceiving virtual and real spaces as similar may ease the transition from physical to digital environments, making it feel natural rather than disruptive. Such beliefs may also buffer against social norms that define "meaningful life" as exclusively offline. Thus, perceived equivalence emerges as a novel predictor of virtual migration, expanding prior models focused on functionality or user needs (Hou et al., 2011; Chang et al., 2014; Atari et al., 2022). Our study bridges real-world and virtual migration research, highlighting analogous decision-making processes in both domains.

However, the direct effect of place attachment on virtual migration was stronger than its indirect effect through perceived equivalence. This indicates that the core driver remains the degree to which individuals' needs are fulfilled in virtual settings. This finding aligns with migration literature showing that people tend to stay in places they are attached to or seek similar environments elsewhere (Feldman, 1996; Low & Altman, 1992), as well as the push–pull–mooring model (Moon, 1995). In our study, both push factors (e.g., dissatisfaction with real-world places) and pull factors (e.g., preference for virtual ones) significantly predicted virtual migration intent.

Future research should investigate for whom the perceived equivalence between real and virtual places serves as a particularly strong predictor of the desire for virtual migration. On the one hand, this may apply to individuals who are sensitive to social norms regarding what constitutes a "meaningful life." For these individuals, normalizing the idea that time spent in virtual spaces is just as valuable as time spent in the physical world could provide an additional reason to transfer their activities to virtual places. On the other hand, people frustrated by their real-world conditions—living in poor circumstances or being unable to meet basic needs—may view virtual spaces as an escape that offers greater opportunities (Deleuze et al., 2019; Liao et al., 2022). For them, the status of virtual places might even surpass that of their unsatisfactory real-world environments.

10.5. SNS and computer game use and perceived equivalence of real and virtual places

The perception of equivalence between real and virtual places can be influenced by various factors beyond attachment to these environments. We expected that the current level of use of virtual places would be a necessary but insufficient condition for developing this belief. Regression analyses confirmed that in both adult samples, social networking site (SNS) use was a weak but significant predictor of perceived equivalence, while games/VR use was not. This suggests that interpersonal interaction plays a key role in shaping perceptions of digital environments as "place-like." These findings are consistent with research on place-making, which shows that social interaction is crucial to transforming spaces into meaningful places (Switalski et al., 2023; Florida, 2002; Johnson et al., 2014).

However, in the adolescent sample, neither the use of social networking sites (SNS) nor engagement in games/VR predicted perceived equivalence. One possible explanation is that younger individuals have already normalized virtual environments as meaningful contexts for social interaction, identity development, and everyday activities (Manago & McKenzie, 2022). Therefore, their perception of equivalence may be shaped by these deeper psychological and developmental factors rather than by the mere frequency of use.

While games often facilitate interactions with other players, our additional analysis—distinguishing between single-player games, multiplayer games, and VR use—found that none of these were significant predictors of perceived equivalence. One interpretation of the significance of SNSs, but not games, in shaping this belief could be the degree to which these virtual environments allow for the experience of reality. Content on SNSs is often closely tied to the real world, from

personal experiences to current and polarizing political issues.By contrast, interactions in games may be more detached from the real world, focusing instead on activities and themes related to the game environment. Consequently, SNSs may provide fewer opportunities for escapism, defined as using virtual places to avoid life difficulties (Demetrovics et al., 2011). Thus, greater similarity between SNSs and the real world in terms of both content and experiences may translate into a stronger perception of equivalence between real and virtual worlds than in the case of games.

We also tested whether SNS and games/VR usage moderated the mediation between place attachment and willingness to migrate virtually via perceived equivalence. Across all three samples, the mediation held regardless of usage level.

Together, these results suggest that while SNS use may slightly influence perceptions of equivalence among adults, its role is limited, and absent among adolescents. Future research should examine how platform-specific features, content, and user practices shape perceived equivalence between virtual and real places.

10.6. Demographic predictors of perceived equivalence

Our results indicate that demographic factors play a limited role in predicting perceived equivalence, with the exception of age in adult samples. Interestingly, the effect of age differed across countries: in the UK, younger adults reported higher perceived equivalence, while in Poland, the opposite pattern emerged. This divergence may reflect different trajectories of technological adoption. In the UK, earlier mainstreaming of digital environments may have fostered greater digital literacy—but also greater skepticism—among older adults. In Poland, where adoption occurred later, older individuals may view virtual spaces with less critical distance. We hypothesize that these discrepancies stem more from national variation in older adults' digital attitudes and skills than from differences among younger people, for whom digital environments are a normative part of life. However, our data do not allow for definitive conclusions about the reasons behind these differences. Future research should explore this further, for example, through large-scale cross-cultural studies examining how perceived equivalence varies across age groups and whether these differences result from distinct technological adoption trajectories and levels of digital literacy (see e.g., Elimelech et al., 2022; Chee, 2024.

11. Limitations

The first important limitation is related to our approach to developing the measure of perceived equivalence. It was guided by a theoretical framework, focusing on capturing the core philosophical concept of perceiving virtual places as equivalent to real-world location. Our measure demonstrated stability across two cultural contexts (Poland and the UK) and age groups (adults and adolescents), showing theoretically expected associations with related variables such as relative place attachment and willingness to migrate to virtual worlds. However, our approach differs from the standard development of questionnaires, which typically involves selecting items from a broad initial pool. Therefore, our scale may not cover all aspects of perceived equivalence. In future research, we plan to expand the item set to examine additional potential dimensions of equivalence, such as perceived quality of life and meaningful experiences in virtual and physical spaces.

Our study relies on self-report measures, which may be subject to biases like social desirability. However, given the anonymity of our survey and the neutral nature of our constructs, this risk is likely minimized. Future research could incorporate behavioral data, such as tracking actual time spent in virtual spaces.

The correlational design limits the ability to establish causality. Future research should include experimental and longitudinal designs to confirm the causal relationships between the examined variables. For example, place attachment could be experimentally manipulated by enhancing a sense of connection to either a virtual or physical location and examining its effect on perceived equivalence. Similarly, perceived equivalence could be manipulated by exposing participants to narratives that emphasize either the seamless integration or distinctiveness of virtual and physical places, providing a clearer understanding of its causal effect on willingness to migrate to virtual spaces.

Although our studies included representative samples from two different countries, broader research across diverse populations would provide a more comprehensive understanding of how different groups perceive the balance between real and virtual places. Future studies should focus on populations with higher levels of technologization (e.g., South Korea and Japan). These groups are likely to provide insights into more nuanced perceptions of the equivalence between real and virtual spaces.

Additionally, while the mediation effects were statistically significant, their magnitude was relatively small. Future research should examine additional variables, such as different types of presence, technological familiarity, or social norms around technology use, to better as predictors of perceived equivalence and willingness to migrate to virtual spaces.

12. Conclusion

The rapid advancement of technology continues to fuel the long-standing philosophical debate about the nature of reality and authentic experiences. While philosophical discussions have extensively explored the distinction and overlap between real and virtual places, empirical research on how people perceive these concepts remains limited. We introduce the concept of perceived equivalence—the idea that some individuals experience virtual spaces as just as real and meaningful as physical locations. Our studies show that only a minority of adult participants believe in the equivalence of real and virtual places, aligning more closely with Nozick's view that virtual experiences are inherently inferior. Conversely, adolescents show a higher level of belief in equivalence, suggesting not only a generational shift in perceptions of virtual environments but also greater resonance with Chalmers' virtual realism, which views virtual environments as capable of providing meaningful experiences. At the same time, both adults and adolescents with stronger attachments to virtual spaces are more likely to view them as equivalent to real-world environments. This belief, in turn, correlates with a greater willingness to shift more aspects of life into virtual worlds. Our findings may inspire further research on integrating philosophical theories about the reality of virtual places with empirical insights into how these spaces are perceived and experienced. They also show the need to examine the broader implications of "virtual migrations" for mental health and social cohesion as younger generations continue to blur the boundaries between online and offline worlds.

CRediT authorship contribution statement

Tomasz Oleksy: Writing – original draft, Visualization, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Izabela Lassota:** Writing – original draft, Validation, Project administration, Formal analysis, Conceptualization.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author(s) used ChatGPT for proofreading purposes. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

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Data availability

Data will be made available on request.

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